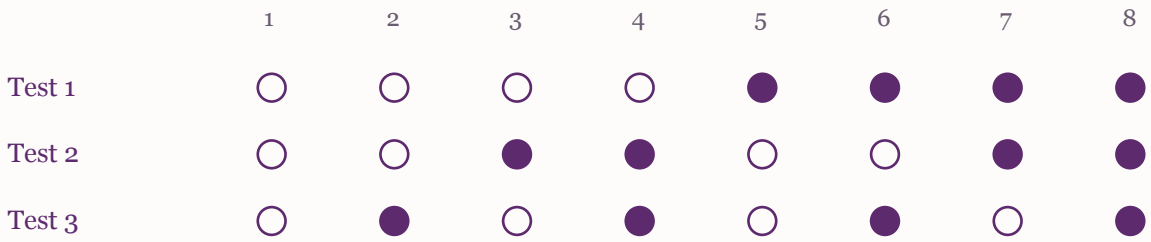
The Independence Property of Model Theory

A Mathematical Rosetta Stone

Olivia Aubone and José Iovino

A limit of continuous functions need not be continuous. Starting from this familiar fact, we follow a restriction on independent patterns through probability, Banach geometry, dynamics, quantum measurement, and numerical computation.

THREE TESTS · EIGHT PRESCRIPTIONS



Each column is one witness. Filled circles mean high; open circles mean low. Every prescription on the three tests occurs.

COMBINATORICS

The tests ask for membership in sets.

ANALYSIS

The tests are values of real functions.

DYNAMICS

The tests observe one orbit at selected times.

For an infinite independent sequence, every prescription on any finite selection of tests must have a witness. Excluding such a sequence leads to a theory of limits whose consequences can be read in several mathematical languages.

Model theory supplies the connections, not a prerequisite.

The main development is accessible without a previous course in model theory. Optional readings give the logical interpretation. Each lecture develops a principal theorem or a concrete application.

COURSE INFORMATION · MEETINGS · ENROLLMENT

utsa-quantum-math.github.io/courses/rosetta-stone/